La Scuola d'Italia / Huson / Physics Nickname:
30 October 2026

2.5 Homework: Mental Math (no calculator)

Work each question in your head — no calculator, and no need to show work. Write just the answer, with its units.

Reference. kilo (k) = 1000 centi (c) = $\frac{1}{100}$ milli (m) = $\frac{1}{1000}$ 1 m/s = 3.6 km/h

1. Fill in the blanks.

   (a) 1.6 km = _____ m

   (b) 85 cm = _____ m

   (c) 0.25 m = _____ cm

   (d) 4 mm = _____ cm

   (e) 2.2 m = _____ mm

   (f) 500 mm = _____ m

2. Squares and square roots.

   (a) 3^2 = _____ (c) 12^2 = _____ (e) 0.5^2 = _____ (g) $\sqrt{81}$ = _____

   (b) 7^2 = _____ (d) 20^2 = _____ (f) 0.3^2 = _____ (h) $\sqrt{144}$ = _____

3. Work out $\frac{1}{2} \times a \times t^2$. Square first, then multiply.

   (a) $\frac{1}{2} \times 10 \times 2^2$ = _____

   (b) $\frac{1}{2} \times 10 \times 3^2$ = _____

   (c) $\frac{1}{2} \times 4 \times 5^2$ = _____

   (d) $\frac{1}{2} \times 0.2 \times 10^2$ = _____

4. Use 1 m/s = 3.6 km/h.

   (a) 10 m/s = _____ km/h

   (b) 20 m/s = _____ km/h

   (c) 108 km/h = _____ m/s

   (d) 360 km/h = _____ m/s

5. Circle the best answer.

   (a) One foot is about how many meters? (A) 0.3 (B) 0.5 (C) 3.28 (D) 12

   (b) One mile is about how many km? (A) 0.6 (B) 1.6 (C) 2.54 (D) 5280

   (c) 60 mi/h is about how many m/s? (A) 16 (B) 27 (C) 60 (D) 97

6. Write each fraction as a decimal.

   (a) $\frac{2}{5} =$ _____ (c) $\frac{3}{8} =$ _____ (e) $\frac{1}{20} =$ _____ (g) $\frac{1}{6} =$ _____

   (b) $\frac{4}{5} =$ _____ (d) $\frac{7}{8} =$ _____ (f) $\frac{1}{25} =$ _____ (h) $\frac{5}{6} =$ _____

7. Fill in the blanks.

   (a) 0.25 hour = _____ min (e) 12 min = _____ h (decimal)

   (b) 0.3 hour = _____ min (f) 20 min = _____ h (fraction)

   (c) 1.5 min = _____ s (g) 1 day = _____ min

   (d) 150 s = _____ min (h) 1.75 hours = _____ h _____ min

8. Physics meets twice a week for 45 minutes.

   (a) How many hours of physics is that each week? _____

   (b) A school year has about 36 weeks of classes. About how many hours of physics is that? _____

9. A 10K race is 10 kilometers long. About how many miles is that? _____

10. A stone is dropped from a tall building. Use $g \approx 10$ m/s^2: each second the stone gains 10 m/s of speed, and after t seconds it has fallen $\frac{1}{2} \times 10 \times t^2$ meters.

    (a) Fill in the table.

t (s)	1	2	3	4	5
speed (m/s)					
distance fallen (m)					

    (b) How many seconds does it take to fall 320 m? _____

    (c) The roof of the Empire State Building is 381 m high. A stone dropped from it lands between which two whole numbers of seconds? _____ and _____ s